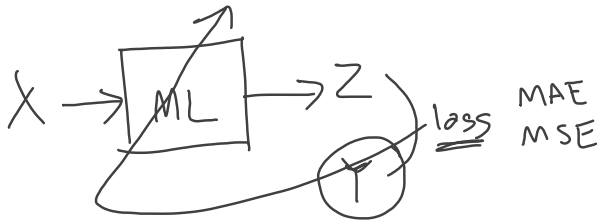
Clustering

Friday, October 02, 2026 8:18 AM

Classification
Regression
} Supervised Learning (X, Y) Labeling (Low)

k-NN

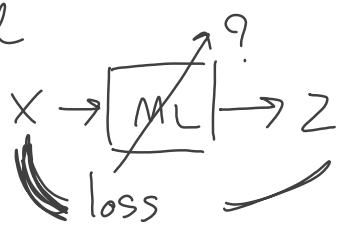


k-means

Clustering - Unsupervised

k Centroids / Cluster

Config



#	A_1	A_2	A_3
x_1	1	1	2
x_2	0	0	1
x_3	5	4	3
x_4	6	3	2

$k = 2 - 4$

① $C_1 = [1 \quad 1 \quad 2]$ ← random

② $C_2 = [0 \quad 0 \quad 1]$ ← random

manhattan

iter 1

D	C_1	C_2	Label
x_1	0	3	① ✓
x_2	3	0	②
x_3	8	11	① ✓
x_4	7	10	① ✓

$C_1 = \frac{x_1 + x_3 + x_4}{3} = [4, 2.5, 2.3]$

$C_2 = x_2 = [0, 0, 1]$

iter 2

D	C_1	C_2	Label
x_1	5	3	②
x_2	8	0	②
x_3	3	11	①

iter 3

$C_1 = [5.5, 3.5, 2.5]$

$C_2 = [0, 0, 1]$

D	c_1	c_2	Label
x_1	7.5	1.5	2
x_2	10.5	1.5	2
x_3	7.5	9.5	1
x_4	7.5	8.5	1

x_4 | 2.6 | 10 | 1

Fuzzy C-means (k-means = Hard C-means)

$\mu^{(m)}$ → fuzziness ≈ 2

		membership
x_1 → c_1	1	0.6
c_2	0	0.3
c_3	0	0.1

hard : Soft

$\sum \mu_i = 1$

$$\sum_{i=1}^{k}\sum_{j=1}^{N} \|c_i - x_j\|$$

Fuzz

D

Soft	c_1	c_2	c_3	
x_1	0.6	0.3	0.1	1
x_2	0.3	0.4	0.3	1
x_3	0.2	0.3	0.5	1
	1.1	1.0	0.9	

hard	c_1	c_2	c_3
x_1	1	0	0
x_2	0	1	0
x_3	0	0	1
	1	1	1

$$J = \sum_{i=1}^{k}\sum_{j=1}^{N} \mu_{ij}^{m} \|c_i$$

$$\sum_{i=1}^{k} \mu$$

$$\frac{\partial J}{\partial c_i} = 0$$